Using policy maps and Shapley explanations to diagnose an RL pricing policy and run targeted learning experiments

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XAI Project 1

The revision closes 36% of the baseline gap to the simulator optimum and increases expected revenue by 1.58% over the baseline.

01 The starting RL policy adapts too little

The hotel simulator uses an assumed demand curve, with 330 days and 10 rooms for one arrival date.

State	Days until arrival and rooms remaining.
Action	Offer 55, 70, 90, 110, or 125.
Reward	Sale price if sold, otherwise 0.
Goal	Maximize revenue before arrival.

Starting point	Before revision
Pricing behavior	**Usually offers 90**
Expected revenue	**851.00**
Simulator optimum	**888.41**
Gap to optimum	**37.41**

Conservative Q-Learning (CQL) is an offline reinforcement learning (RL) method that learns price scores from saved sales and no-sales.

02 We targeted learning near arrival

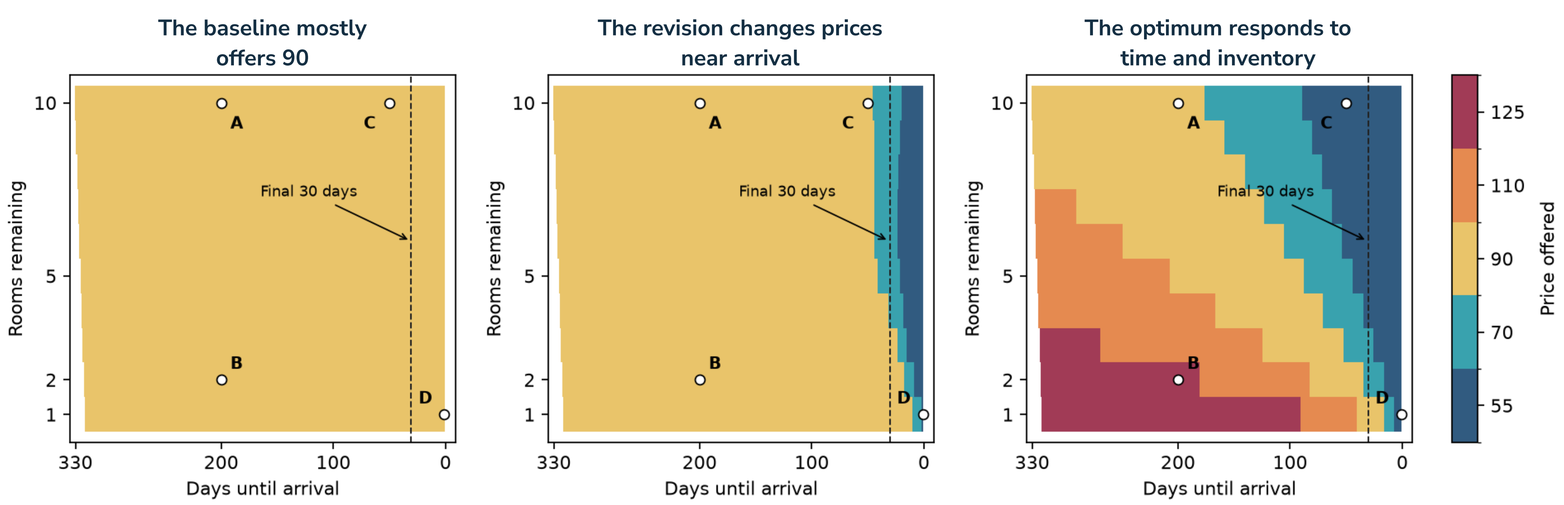
Policy maps and Shapley explanations showed weak deadline adaptation.
Only 22 of 400 added selling periods reached the final 30 days.

New experience	Add 150,000 fresh outcomes covering all states and prices in the final 30 days.
Late replay	Sample late records more often during training.
Time input	Test logarithmic time to emphasize differences near arrival.
Training	Increase updates from 20,000 to 100,000.

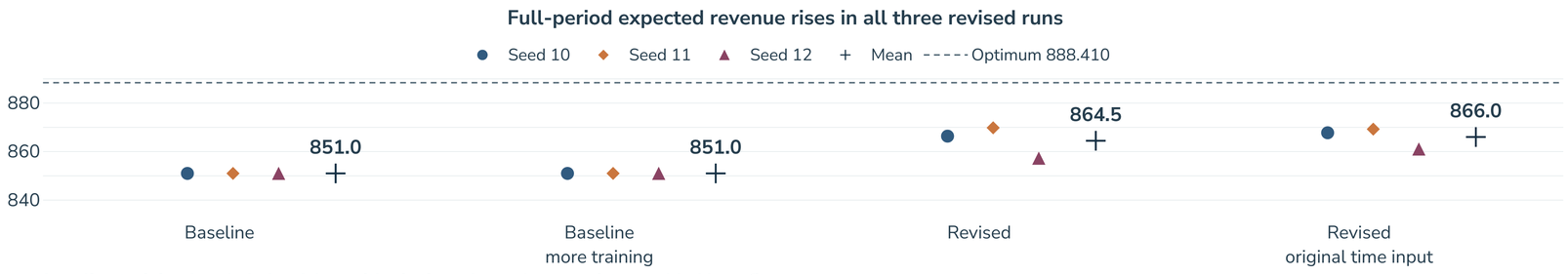
We collected new outcomes, then froze the data for offline training.

03 Revenue improves, but scarce rooms remain underpriced

These maps show all reachable states. The model examples use seed 10.



A More time, many rooms: 200 days, 10 rooms
B More time, few rooms: 200 days, 2 rooms
C Less time, many rooms: 50 days, 10 rooms
D Last day, one room: 1 day, 1 room



Longer baseline training barely helped. Logarithmic time showed no consistent added benefit.

Revised runs share one new dataset. The original-time control changes only the time input.

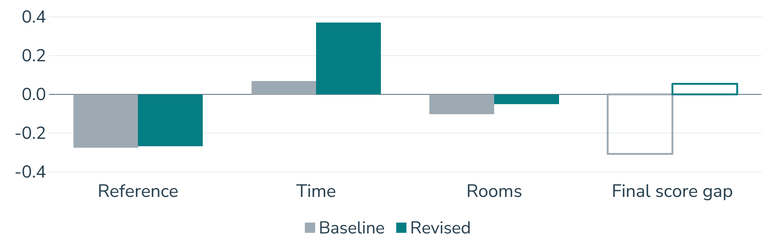
04 Shapley explains progress and errors

Each chart explains a price preference over 90.
Both examples use seed 10.

Reference + time + rooms = final score gap.
Positive favors the alternative price over 90.

D The revised model discounts the last day

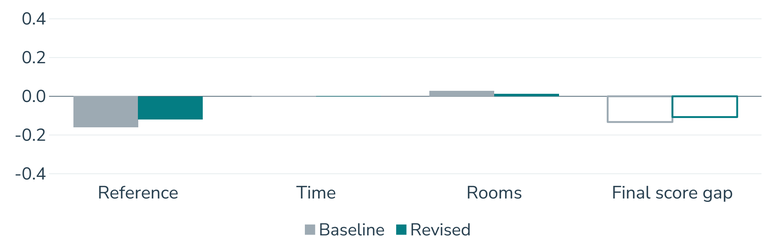
55 versus 90. The simulator favors 55.



A stronger time contribution supports the discount.

B Both models still underprice scarce rooms

125 versus 90. The simulator favors 125.



Both models still choose 90 when 125 is best.